

Supplementary Materials for **Ultrafast rotation of magnetically levitated macroscopic steel spheres**

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Supplementary Materials

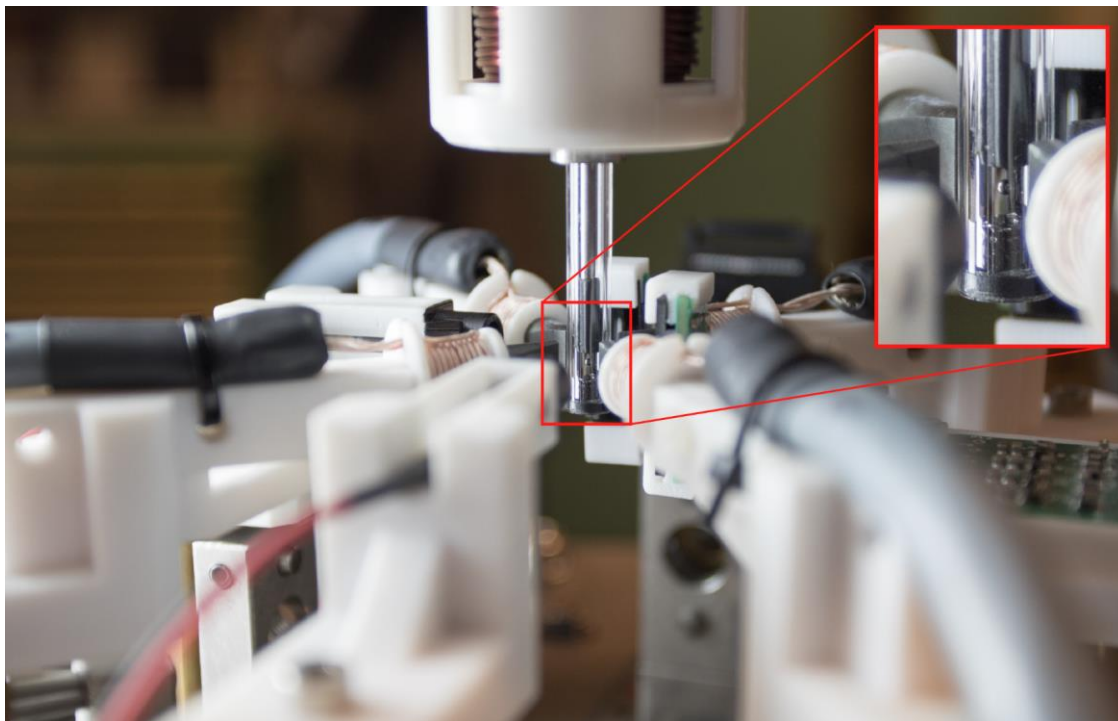


fig. S1. Photograph of the experimental setup with levitated rotor. All relevant motor and sensor components including a zoomed view of the levitated rotor inside the glass vacuum tube are shown.

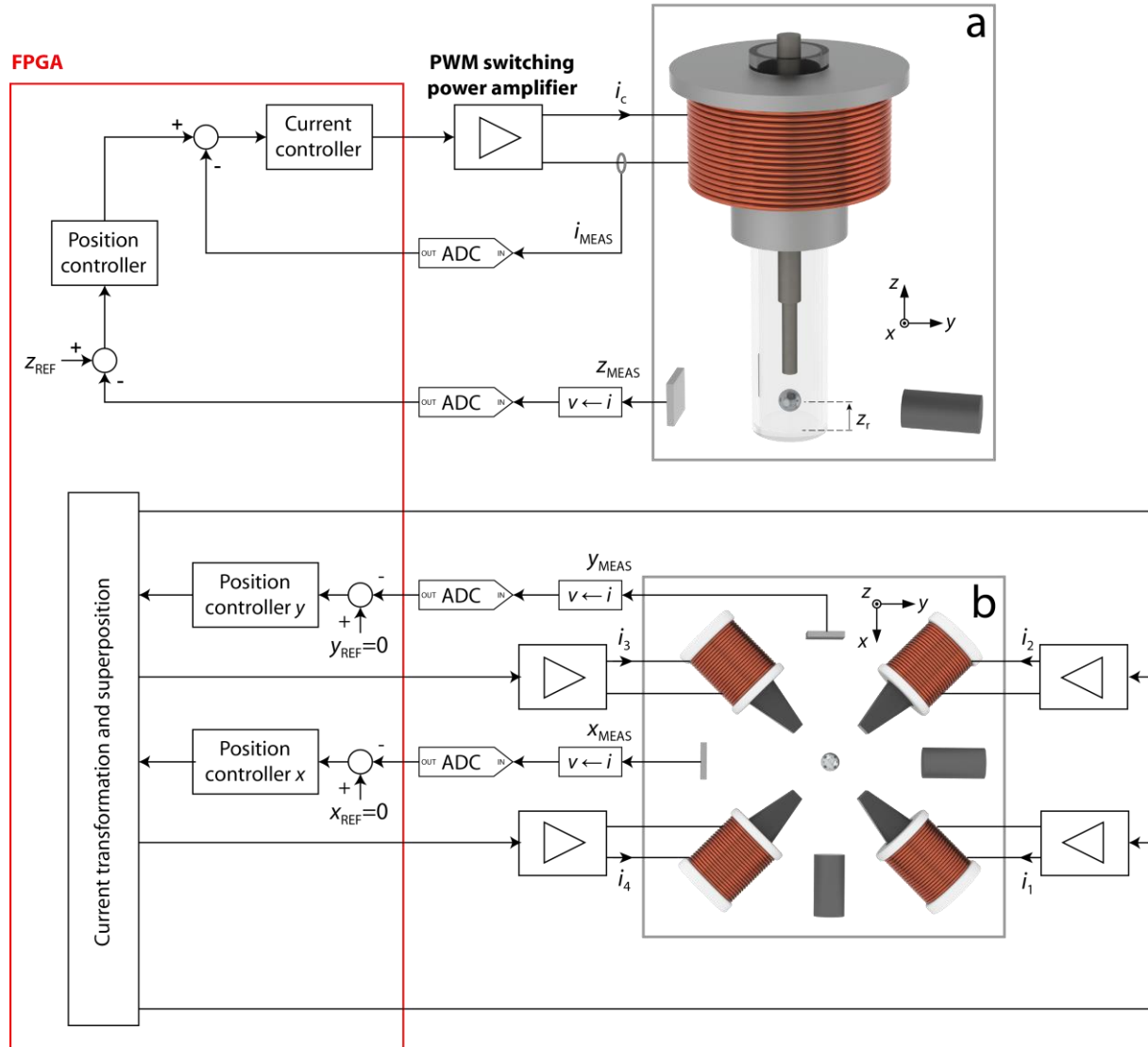


fig. S2. Control structure of the magnetic suspension system. (a) Digital cascaded control for vertical stabilization of the rotor. (b) Direct control of the coil currents by the position controllers without an additional current controller for horizontal stabilization. All controllers are implemented using an FPGA, which also outputs the switching signals necessary for drive field generation.

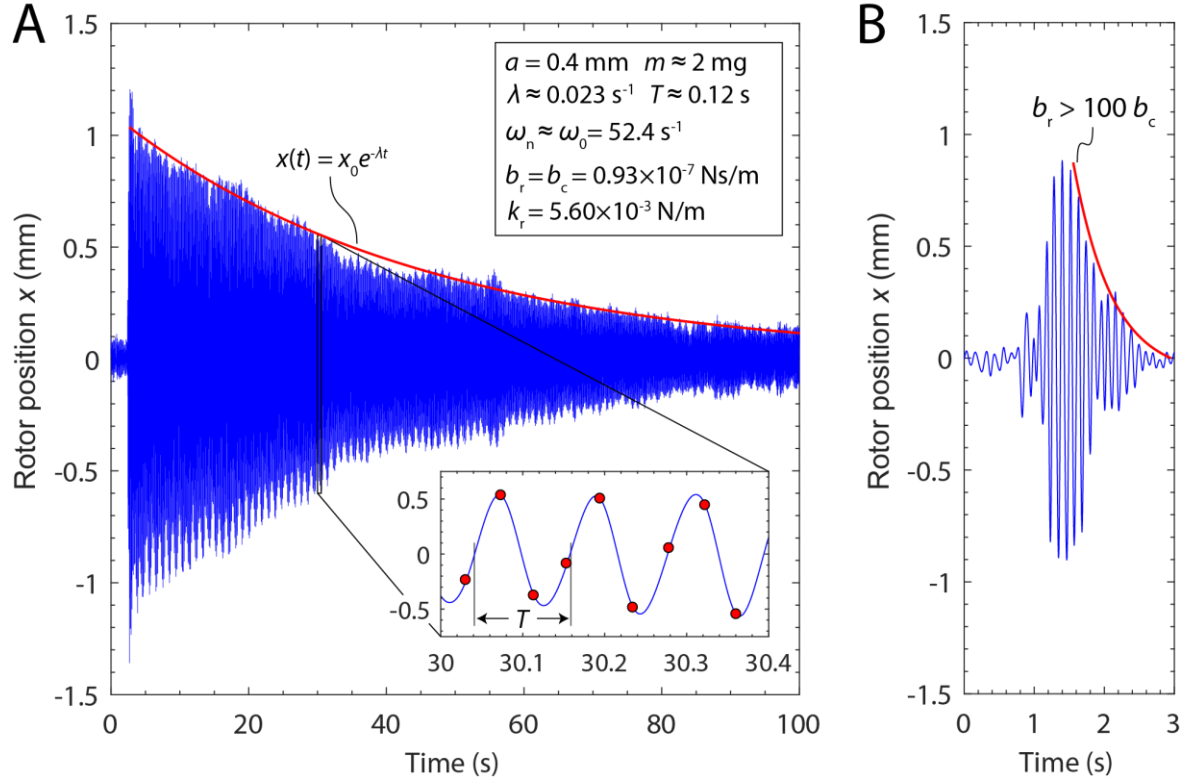


fig. S3. Rotor dynamics in horizontal direction with and without active magnetic damping. (A) Slow exponential decay of harmonic oscillations as a response to a rotor position disturbance due to weak gas friction damping b_c under ambient pressure conditions. The inset shows a zoomed view to determine the frequency of the oscillations. Outlined values were calculated from the measurement. Under vacuum conditions damping is lowered further and reliable stabilization of the rotor is not possible without additional external damping. (B) Oscillations caused by the same disturbance subside within $< 2 \text{ s}$ for the actively damped system, thus significantly improving stability of the magnetic suspension. The radial damping b_r is increased by a factor of more than 100 compared to gas friction damping.